Propositions for Enabling Participation in Performance-Driven Design

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In Performance-Driven Design, it is challenging for different stakeholders such as the end-users to participate in the co-designing process. Performance-driven design requires complex algorithmic calculations, simulations, and optimizations. These computational functionalities enabled for this design process lack of transparency and can be sometimes complicated to understand. Therefore, the current applications of Performance-Driven Design contradict the participatory design where social interactions are considered as important steps to produce desirable and accepted design outcomes.
In this context, the main aim of this study reported in this paper based on a 4-years PhD thesis at Luxembourg Institute of Science and Technology and KU Leuven, is to address research methods suitable for enabling higher levels of participation in Performance-Driven Design and thus to provide recommendations and guidelines.

Keywords: *Performance-Driven Design, Participation Design Process, Architectural Design, Performance and Requirements Modeling.*

INTRODUCTION

Nowadays, a wide range of technologies are employed for the design process in the built environments. These technologies have a significant impact on the process helping in assessing the performance of the design products in terms of aesthetics, comfort, energy, and cost (Chronis, Liapi and Sibetheros, 2012; Shi and Yang, 2013; Abdelalim and O'brien, 2017). The architectural design process has integrated more complex functionalities assisting the design teams to manage information and answer environmental, structural, and other challenges (Monedero, 2000; Frazer *et al.*, 2002; Woodbury, 2010; Haliburton *et al.*, 2011; Park *et al.*, 2013). This shift in practices also reflects Performance-Driven design approach, through which the design is evaluated by different tools providing feedback on its performance to support the decision-making process (Shi and Yang, 2013).

Performance-Driven Design process supports the architects in a more effective and automated process enabled by the integration of optimization methods aiming to inform the design output in relation with a wide set of indicators. It also plays an important role in providing satisfactory outputs for the different stakeholders involved in the design project. However, the level of satisfaction of the end-users remains unguaranteed as pointed by Preiser in his research for the performance evaluation of the office building design (Preiser and Schramm, 2002).

PARTICIPATORY DESIGN AND PERFORMANCE-DRIVEN DESIGN

There are different meanings to the term participation. Despite the common belief about the ideological paradigm of the participation (Agarwal, 2001), the opinions about how to define the process of participation, the people to engage and what is

expected from them remain a complex task (Lyons, Smuts and Stephens, 2001).

The importance of participation in the design process has been the subject of different research and is well established in the literature. Al-Kodmany, (Al-Kodmany, 1999) recognized different benefits from participation, stating that participation gives a stronger sense of commitment, and increases the users' satisfaction regarding the design solution. In addition, sharing additional viewpoints and ideas leads to better facilitation of actions (Price and Mylius, 1991; Kelly, 2001). Regardless the benefits cited previously, there are some obstacles that can be encountered with the participation including time consumption, increase of the financial costs, no guarantee of successful result, professional superiority, uncertainty about the results of public involvement, lack of transparency and empowering specific part of participants (Cleaver, 1999; Claridge, 2004; Miessen, 2017; Mubita, Libati and Mulonda, 2017).

Performance-Driven Design (Figure 1) is consistently progressing on the technological level. The integrated capabilities and functions that can be used during this design workflow, facilitate different tasks used to be complicated in traditional design processes. Computer simulation processes are being integrated into architectural design tools (Attia *et al.*, 2013; Davis, 2013; Wortmann and Schroepfer, 2019), in particular, the computer-aided parametric tools are used to generate, simulate, visualize, and optimize architectural outputs (Camporeale, 2013; Yang *et al.*, 2015; Wen, Daher and Kubicki, 2020). These integrated capabilities -used together or separated- allow the possibility for a customization following the needs.

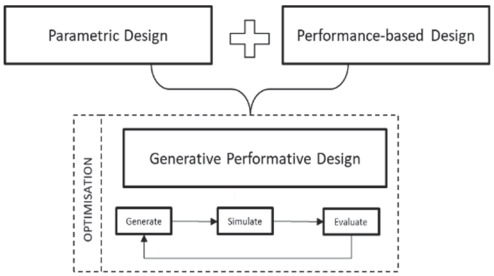


Figure 1
Key factors for participation thought a Performance-Driven design

At the technological level in architectural practices, parametric design and Performance-Driven Design have emerged as novel attitudes toward architecture and urban design. As a result, we are observing novel practical shifts taking place in architectural and urban design with the introduction of Performance-Driven Design process (Oxman, 2017; Gallo and Pellitteri, 2018; Wortmann, 2019). However, Performance-Driven Design has some challenges related to the computer-aided parametric tools used by the designers and the functioning of these tools (Davis, 2013). These experts need to define a modeling workflow and understand the project to be able to identify the different components of the parametric model (Zarei, 2012). From another point of view, Performance-Driven Design, which is based on optimization techniques, encourage a top-down design approach, creating a challenge related to the involvement and participation of the end-users in the different stages of Performance-Driven Design (Daher and Kubicki, 2020).

HYPOTHESIS

For an effective and continuous participation between professionals and end-users in Performance-Driven Design process, there is a need for a digitally supported platform enabling an interaction with the designed model and an exchange of the information. We propose the requirements for a participatory theoretical process that can be enabled by Performance-Driven Design process through a wide range of components supporting the participation.

The key factors that make Performance-Driven Design process a suitable environment for participation are:

- The ability to create a design space where the designers can accommodate changes and modifications with less time (Woodbury, 2010; Davis, 2013).

- The possibility to streamline the workflow from complex geometry to analysis and simulation providing a transparent and trustable design process.
- The easiness in use enabled via the visual programming language allowing the exploration of the solutions (Davis, 2013).
- The decreasing of the time spent for the creation and modification of the design proposals (Daher and Kubicki, 2020).

RESEARCH METHODOLOGY

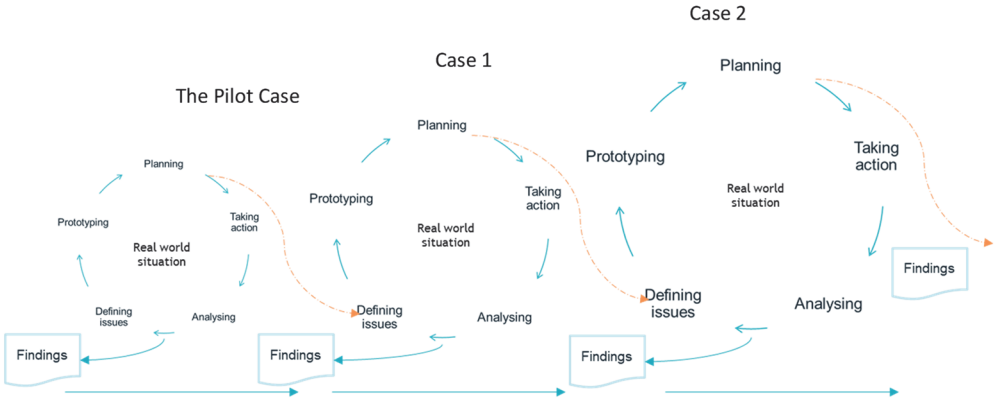
To address this research and evaluate the framework that has been developed during this research project, we explored different design process situations through three case studies focusing on understanding the dynamics of each case study to conclude with recommendations and propositions for enabling participation in Performance-Driven Design process. We apply a design-driven action research which has an objective of producing "research-informed change" to address issues and problems in real life. These three case studies, based on real-life situation, are addressed in this research. For each case study, we conducted several iterations to 1) test and improve Performance-Driven Participatory design process, 2) solve the issues and problems related to output solutions (project). These case studies are:

- The pilot case study deals with the participation in the architectural design of refugee settlement;
- The first case study addresses the early design stage of urban planning;
- The second case study involves the design of interior layout of office building.

RESEARCH TECHNIQUES AND TOOLS

For the development of our case studies and the implementation of the framework, it was essential to identify a set of techniques and tools to construct this framework. The choice of these tools and techniques was based on their ability to simulate different performance indicators. Grasshopper, as a Performance-Driven design modeling tool, was used due to the wide range of functionalities that it included with the possibility to add different plugins. To implement our framework, we have developed an application representing each step of the design process with a tangible user interface creating a connection with Grasshopper thanks to Firefly plugin, and other interactions of the design framework were developed based on different programming language (the system prototype in this thesis is based on TULIP framework developed in Luxembourg Institute of Science and Technology and which is a JAVA software framework for developing tangible applications). This programming is not the focus of this paper, however,

Figure 2
Practice-based design process



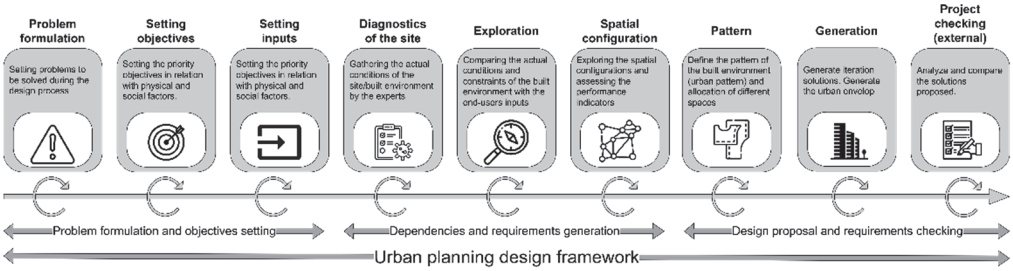


Figure 3
Performance-Driven design process

it was essential to develop this framework to be able to test it with the different case studies.

The tangible tabletop interface is chosen for its ability to gather the necessary information (Wagner *et al.*, 2009) (Figure 4). A key factor of the success of the participatory design process is the easiness of use for the interface and the creation of a transparent design process where the involvement of the end-users and the information exchanges are possible throughout this interface.

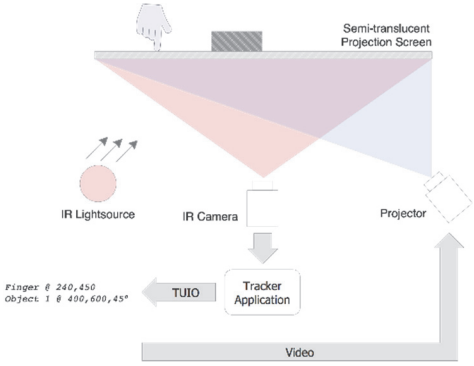


Figure 4
The tangible table interface

PERFORMANCE DRIVEN PARTICIPATORY DESIGN PROCESS

Framework of Performance Driven Participatory design

Setting up a Performance-Driven Participatory design framework to encourage a bottom-up process needs to reflect different design interactions related to any design process. By developing the three case studies, these interactions were identified and grouped into three main interactions. In this paper, we will briefly describe these three interactions (Figure 3).

The first interaction of the framework helps to integrate the end-user's requirements in the design process. These requirements can be defined by consultation or by direct participation. The framework will transform these requirements into digital input that will influence the design in the next interactions.

The second interaction is related to the diagnostic of the site and exploration of the fulfillment between the actual conditions and end-users' requirements.

The third interaction deals with the development of early design solutions based on the requirements of the end-users. It also enables the generation of alternative solutions where the end-users have the possibility to explore and understand the solutions that have been proposed and to compare between these solutions to choose one of them.

Infrastructure supporting the framework

Analysis engines based on computational techniques and optimization modules that allow the recommendation based on the different set of information. Each case study had a specific recommendation component developed with Python language and connected to the main system to enable specific applications for each case study.

Figure 5
The digital set used to enable the participation in Performance-Driven design process

The interface. The challenge is to ensure and facilitate the participation in the design process.Therefore, the interface to use in the framework has an important role to gather heterogeneous data and information related to the physical constraints and end-users' requirements. The input will help later in the design stages to allow the generation and exploration of the design proposals. The tangible table (Figure 5) allows the communication, visualization, and the exchange of the information. This interface is foreseen as a participatory technology that helps to increase the engagement of end-users and community in urban planning (Wagner et al., 2009).

Performance indicators to be considered in the cases studies

Setting the objectives are defined regarding the aim of the design project or some constraints to respect during the design process (e.g., budget, surface…). The aspects that are evaluated during the design process are different depending on each use case.

In general, we considered aspects related to the visual comfort, the well-being (noise, pollution), and the mobility in relation to the integration of the different functions and the proximity between them.

The principal interaction with the end-users starts with the objective setting, considered as an important asset to ensure this continuous connection between the participation process and Performance-Driven design process.

FEEDBACK ON RESEARCH QUESTION AND AIMS

Learning from the case studies

Refugee camp. The participation in humanitarian works is perceived as important factor to ensure the satisfaction of users in their new settlements. For this case study, we presented a method to reach an engagement of the end-users (refugees) in the early design stage of the planification. This case study was not tested with real conditions, however, to ensure an effective engagement of the refugees, we adapted the generic design framework to this case study and developed a technological process integrating social and physical constraints to help in the decision process.

Urban planning. This case study has significantly helped in the progress of the development of the framework of Performance-Driven participatory design process. It also showed different levels of participation in the urban planning process: 1) problem formulation and objectives setting, and 2) dependencies and requirements generation. This design process allowed a direct participation in the two first interactions and enabled a transparency in checking the proposed solutions.

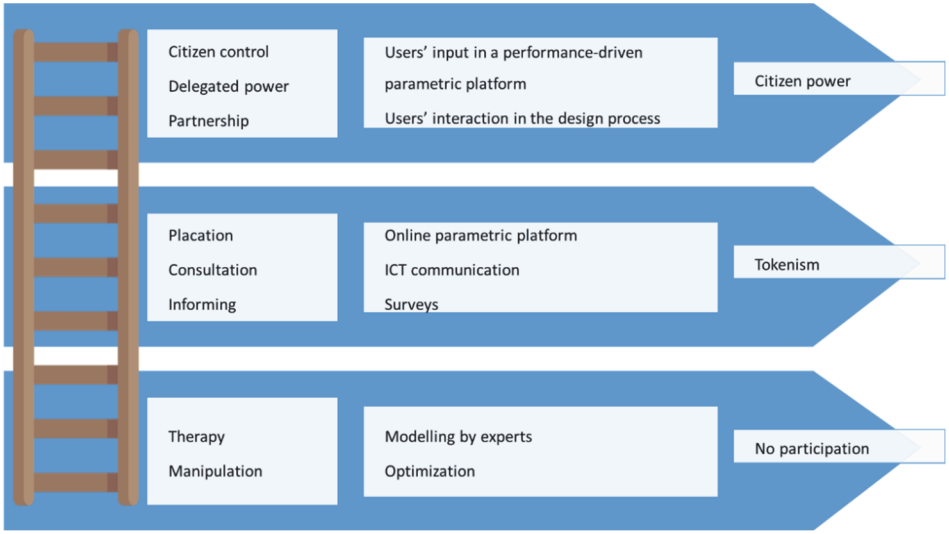


Figure 6
A Ladder of Participation for Performance-Driven Design Participatory process based on the Ladder Of Citizen Participation Arnstein (Arnstein, 1969)

Office layout. The third case study was dealing with the design of the interior layout of an office building. The generic design process was also adapted to answer the context of this case study. After the experiments with the end-users (employees), it was shown that this design process helped them to more understand the space planning considering a set of constraints: 1) the cost of the solution proposed, 2) the association of teams, 3) the quietness and 4) the view.

Conclusion on Performance-Driven Participative design Framework

To enable an integration of a participatory process in a Performance-Driven Design process, we propose a participatory ladder based on the participation ladder of Arnstein mapping the level of participation with recommendations for the technological tasks to enable each level (Figure 6). We have identified main findings:

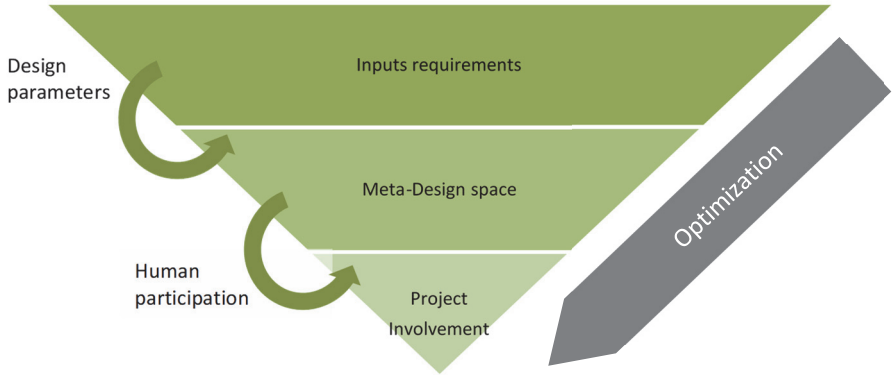
- Each case study is different, therefore, adapting the proposed framework to reflect the specificities of the case study is an important key element. The performance indicators and the requirements of the end-users are unique depending on each case study.
- Conducting different meetings with end-users should strive for a group consensus, and the decisions had to be made with compromises.
- The inclusion of the human perspective is essential factor in the design that enables to engage external requirements encouraging a diversity in the points of view.

RECOMMENDATIONS FOR AN EFFECTIVE PERFORMANCE-DRIVEN PARTICIPATORY DESIGN FRAMEWORK

The use of interactive tools and visualizations techniques

Keeping a transparent relation between the input and the solutions can ensure a participatory approach in a Performance-Driven design process. However, there is an essential key factor to allow better interaction and visualization of the information. The end-users will be able to find the focus and the most important aspects to be considered by the designers and be actively engaged in the design process.

Figure 7
Role of optimization in Performance-Driven Participative design framework



Ease of use and user-friendly interactions are considered as central features to enable a participation in a digital design framework. Therefore, the system includes an interface to manage the input in a simple and easy way and screens to evaluate and visualize the results of the iterations.

The design of a meta-model

The creation of a meta-model enables the design of a focused "solution space" and iterations that satisfy the needs and requirements of the end-users. The benefit of meta-model design is in the ability given to accommodate changes and modifications in the design that can be requested by the end-users with less time and effort and to give the end-users the ability to access and investigate these solutions.

Exchange information

In a Performance-Driven design process, there is a need for the end-users to contribute to the design process and receive clear information. Sharing the knowledge between the different actors in the design process is important to provide better solutions. It is important to create a smooth exchange of information that can enable an engaging communication and discussion between the experts and the end-users.

Transparency

As mentioned previously, exchanging information is a key factor to allow a participatory process in a Performance-Driven design process. However, the framework showed that there was an easy understanding by the end-users of the information exchanged. This was due to the use of simplified information where the end-users, considered as non-experts, were able to better understand the showcased information.

Flexibility

Having a flexible design process helps in reducing the top-down design approach and offers an easiness in proposing alternatives in the design. In this framework, the flexibility of the design generation was ensured by the capacity of providing the input by the end-users through a participative process and modifying these inputs through a participatory approach around a tangible tabletop interface increasing the involvement of the end-users in Performance-Driven Design process.

Adaptation to different design steps

There is an important aspect to include different level of participation in the various design steps. In the developed framework, we identified two main levels of participation (Figure 6). On the conceptual level of the design, we noticed that there is an

importance to include a direct participation with a high level of participation such as the delegation of power. As a recommendation, including the participants in early design stage by defining the requirements, can offer a more value to the design process and helps in identification the important issues to be considered in the design process.

DISCUSSION: THE ROLE OF OPTIMIZATION IN ENABLING THE PARTICIPATION

The design optimization is preferred to be as medium for reflection helping the participants to better understand the outcomes and allow them to examine possibilities with interactive methods and systems. By linking the design solutions to the performance indicators, the participation is enabled by allowing the exploration of the design alternatives in early design stages by the end-users. This helps in making better-informed decision by refining the requirements, comparing different solutions, and exploring the performance of each simultaneously.

The developed Performance-Driven participatory design framework aims to break the usual participation process with the top-down and autopilot process. It presents a theoretical and technological framework for including end-users' feedback and criteria in the urban planning process while also enabling the integration of social criteria, where the end-users are able to input, adjust and define their own preferences and requirements in a participative approach.

The role of optimization in generating design and how to ensure that esthetic requirements are met is questioned. The generation of optimized design solutions should be set according to some design strategies. This means that designers should define design strategies and directions to ensure that the final solutions will meet the esthetic requirements.

From another side, the role of the designer in a performance-driven parametric process is beyond a normal role of designing but also, he/she has a role in infrastructuring the design process on performance-driven process basics. Infrastructuring and including the participation in a Performance-Driven Design process would lead to some abstraction to be made. Ambitions towards using the system to conduct projects is strengthened by the complexity of the design/design process and the intention to include participation in the process.

The outcomes with the use of the system require a connection between the output and participatory process to set the input. The system ensures the outcomes to consider the input. However, participatory approach in Performance-Driven Design process requires major development efforts, including validating and testing the approach to reach the maturity on the best practices and market acceptance for participation.

Therefore, there is a fundamental need that the education evolves and that designers should gain knowledge in computing techniques allowing experts to be more aware about environmental simulations, and participatory procedures helping experts to gain knowledge in the methods and process for participation.

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